Sweatshop on P-Chess: three tickets from Bruno's QA

Held piece survives Maia's reply (UI-011), human reaction time for Maia (BOT-009) and a whole-second clock calibration (BOT-010); GPT-6.1 Sol lineup, lane `qa`

Foreman, session `sweatshop/qa/2026-10-27-1653`

27 October 2026

Summary

A lane run (`-Lane qa`) in a fresh clone at `.artifacts-build/lanes/qa`, default lineup `sol`. All three tickets are `Difficulty: hard`, so stage 2 ran on the lineup's hard model, `gpt-6.1-sol xhigh`, the same model as the reviewer. Before launch the clone's gate passed (258 passed, 0 failed) and the ignored Lc0 binary and Maia nets were copied in from the main checkout. Launch at 16:53, "Nothing runnable" at 17:44. UI-009 (blocked on sound research in another session) and CI-003 (blocked on Bruno's clean-Windows game) were out of scope and untouched.

- **Produced:** UI-011, BOT-009 and BOT-010 merged on the session at the first attempt, `Verdict: Approve`, 0 findings each. BOT-010 is `Review: human`. Session PR #22; gate 268 passed, 0 failed (258 on `main` before the run); Python calibration tests 14 passed.
- **Cost: $5.86** at API list price over 6 stages, all `gpt-6.1-sol xhigh`: $3.55 stage 2, $2.30 review. Codex quota: 2 points of the weekly window (stage 2 1, review 1, each ±1).
- **What went wrong:** nothing in the run. Every ticket still owes a manual game (product acceptance pending).
- **Provisional verdict:** clean and fast (17 min per ticket end to end, $1.95 per ticket). Hard tickets on `xhigh` cost about twice the `high` stage 2 of 25 October.

What was done

UI-011: when you hold a piece (dragging or click-selected) while Maia thinks and Maia moves, the hold now continues and the drop counts against the new position, as on Lichess and Chess.com. If Maia captured or moved the held piece, the hold ends; legal-move dots refresh; an open promotion picker still closes. Main file `ui/board_view.gd` (`_commit`).

BOT-009: Maia no longer answers instantly to a hung piece. An ordinary move now waits at least 0.4 s (`REACTION`, an initial value to tune by playing), so Maia's clock can reach zero in a 1+0 scramble. The 0.05 s "premove" floor now needs the human's last move to have been Maia's top prediction, top_p over 0.9 and under 10% of the starting time; Maia asks Lc0 for that prediction with one extra policy query. Main files `bot/time_model.gd` and `bot/maia_player.gd`.

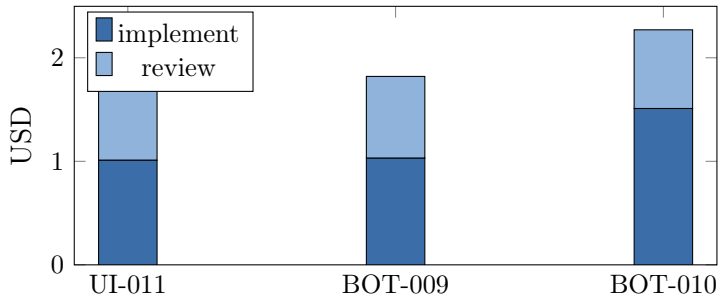
BOT-010: the think-time table is recalibrated. The script now fits the log-normal to Lichess's whole-second clocks (interval-censored likelihood) and samples up to 1,000 positions per (control, level, clock bin), so the 1+0 scramble buckets have data: 243 buckets, 216–479 samples in each of the 27 scramble buckets, from a 20 MB prefix of the September 2026 dump. Main files `scripts/calibrate_time.py` and `data/time_model.json`.

Timeline

16:51	Clone of `origin` at `bc36ac9`; gate 258/0; engine files copied.
16:53	Launch, pid 29500; session `sweatshop/qa/2026-10-27-1653`. UI-011 stage 2.
17:05	UI-011 to review.
17:08	UI-011 merged, 0 findings. BOT-009 stage 2.
17:19	BOT-009 to review.
17:22	BOT-009 merged, 0 findings. BOT-010 stage 2.
17:32	BOT-010: two proxy decisions (GPT-6 Astra fallback) on the download size.
17:39	BOT-010 to review.
17:44	BOT-010 merged, 0 findings; "Nothing runnable"; PR #22 opened.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
17:05	UI-011	impl	sol xhigh	12	3.1M	97%	22k	1.01	to-review
17:08	UI-011	review	sol xhigh	3	1.7M	89%	9k	0.75	merged
17:19	BOT-009	impl	sol xhigh	11	3.1M	97%	25k	1.03	to-review
17:22	BOT-009	review	sol xhigh	3	2.1M	93%	10k	0.79	merged
17:39	BOT-010	impl	sol xhigh	17	5.2M	98%	28k	1.51	to-review
17:44	BOT-010	review	sol xhigh	4	2.1M	94%	12k	0.76	merged
				50	17.3M		106k	5.86	



Wasted: none in the driver. Inside BOT-010's stage 2, a first calibration attempt with a 500 MB cap was stopped before producing a table (parsing projected about two hours); its download is estimated at 200 MB, within Bruno's 500 MB approval.

Comparison with previous runs

From `run-log.md`: the 25 October `sol` run cost $4.18 for three tickets with stage 2 on `gpt-6.1-sol high` ($0.43–0.98 per stage 2). This run's stage 2 on `xhigh` cost $1.01–1.51, for hard tickets; reviews cost the same ($0.57–0.91 then, $0.75–0.79 now). BOT-010 is the most expensive ticket ($2.27): stage 2 ran a real calibration over Lichess data.

Observations per model

- **gpt-6.1-sol xhigh, stage 2**: three handoffs at the first attempt, 11–17 min, each with red test commits before the fix and a final red replay against the old production file. UI-011 changed one existing test's expectation, as its ticket asked. BOT-009 edited three existing test files only to feed the new prediction reply (no assertion weakened). BOT-010 loosened one CLI assertion (exact median → between 2 and 4 s) because the rounded-clock fit no longer equals the raw median; the per-stratum bound is tested by a separate test.

- **gpt-6.1-sol xhigh, review**: three reviews, 3–4 min, 0 findings, no code in review (no round-0 rows), all reusing the stage-2 gate receipt. The BOT-010 review noted and accepted missing commit bodies as a Standards exception.

Driver and runtime issues

1. None in the driver: no crash, retry or restart; shutdown 1.3–1.8 s per stage.
2. `.artifacts-build/` was not ignored in the main checkout; the foreman added it to `.git/info/exclude`, and the clone's exclude got the run-log paths (excludes are not cloned).
3. A fresh clone lacks the ignored `engine/lc0` and `engine/weights`; without them four Lc0 tests skip. The foreman copied them from the main checkout.

Recommendations

1. Play the three manual checks below before tagging a release; BOT-009's 0.4 s and BOT-010's table are tuned by feel, not by tests.
2. Lane clones: document the engine copy as a clone setup step next to the excludes.
3. Keep `gpt-6.1-sol xhigh` for hard tickets: scoreboard row `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh`, 9 reviews, 1 reopened (11%), S2 rate 11% — under 10 reviews, an anecdote.

Human debt and proxy decisions

No `Débito humano:` lines. Proxy decisions of this run, both on BOT-010 by the GPT-6 Astra fallback:

- "start with a 20 MB compressed prefix and increase only if coverage is insufficient, within the cumulative 500 MB download budget; acceptance still requires all 27 real 1+0/clock_bin 0 buckets with at least 50 samples each."
- "reserve an estimated 200 MB for the interrupted attempt and proceed with the 20 MB prefix. Including the earlier 20 MB probe, conservative accounting reserves 240 MB of 500 MB and leaves 260 MB."

Reopen attribution

No reopened review and no fix made in review in this run, so no new rows in `reopen-attribution.md`. No session-review, release-audit or post-release finding was reported since the last pass (the Copilot reviewer on PR #22 did not run, out of quota).

Scoreboard

`scoreboard.ps1` over the ClassyCall, EasyMath, P-Chess, PhysiSyst, SIGAA-ME and SynchroNice main checkouts (no lane clone). Moved since the 25 October report: `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh` from 6 to 9 reviews, still 1 reopened (17% → 11%), an anecdote. `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` unchanged at 18 reviews, 22%. The implementer row `gpt-6.1-sol xhigh`: 9 stage runs, 9 to review, 0 failed, median 12 min. Interventions printed before this run's two proxy rows were committed: human-rescue 24, routine-approval 18, proxy-decision (gpt-6-astra high, fallback) 44, (opus) 33.

Pending for the human

1. UI-011: a 1+0 game; start dragging before Maia replies and drop after; the move plays.
2. BOT-009: a 1+0 game; hang a piece with Maia under 6 s (it does not move instantly); a premove-only scramble where Maia's clock reaches zero. Tune `REACTION` if needed.

3. BOT-010 (`Review: human`): a few 1+0 and 5+0 games against the new table; judge the think-time spread.
4. Review and merge session PR #22.